

**COURSE TITLE : SPECIALIZED PRINTING APPLICATION**  
**COURSE CODE : 6103**  
**COURSE CATEGORY : A**  
**PERIODS/ WEEK : 5**  
**PERIODS/ SEMESTER : 75**  
**CREDIT : 5**

**TIME SCHEDULE**

| MODULE       | TOPIC                                             | PERIODS   |
|--------------|---------------------------------------------------|-----------|
| 1            | Waterless offset, dry offset and print on demand. | 15        |
| 2            | Textile printing and Printed electronics          | 15        |
| 3            | 3D printing and Nanography                        | 20        |
| 4            | Lenticular printing and Hologram                  | 25        |
| <b>TOTAL</b> |                                                   | <b>75</b> |

**Course Outcomes:**

| SUB | ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE : |
|-----|----------------------------------------------------------------------|
| 1   | To comprehend waterless offset                                       |
| 2   | To understand dry offset printing                                    |
| 3   | To understand Print on demand                                        |
| 4   | To comprehend textile printing techniques                            |
| 5   | To understand about Printed electronics                              |
| 6   | To know 3D printing                                                  |
| 7   | To know Nanography                                                   |
| 8   | To understand Lenticular printing                                    |
| 9   | To understand Hologram                                               |
| 10  | To know hybrid printing                                              |

**MODULE I WATERLESS OFFSET, DRY OFFSET AND PRINT ON DEMAND.**

**1.1.0 To comprehend waterless offset**

- 1.1.1. To explain principle of waterless printing
- 1.1.2. To explain about waterless plates
- 1.1.3. To specify the merits and demerits of waterless printing

**1.2.0 To understand dry offset printing**

- 1.2.1. To explain dry offset
- 1.2.2. To specify the dry offset plates
- 1.2.3. To differentiate dry offset and waterless printing

### **1.3.0 To understand Print on demand**

- 1.3.1. To explain print on demand
- 1.3.2. To specify the advantages and disadvantages of print on demand

## **MODULE II TEXTILE PRINTING AND PRINTED ELECTRONICS**

### **2.1.0 To comprehend textile printing techniques**

- 2.1.1 To explain textile printing
- 2.1.2 To describe methods of textile printing
- 2.1.3 To explain digital textile printing
- 2.1.4 To explain textile printing machine

### **2.2.0 To understand about Printed electronics**

- 2.2.1 To explain printed electronics
- 2.2.2 To define printed electronics production methods
- 2.2.3 To specify the printed electronics advantages
- 2.2.4 To describe printed electronics production methods

## **MODULE III LENTICULAR PRINTING AND HOLOGRAM**

### **3.1.0 To know 3D printing**

- 3.1.1 To explain 3D printing process
- 3.1.2 To describe different 3D printing processes
- 3.1.3 To explain different application of 3D printing
- 3.1.4 To define Bio-printing and nano scale 3D printing

### **3.2.0 To know Nanography**

- 3.2.1 To explain Nanography printing technology
- 3.2.2 To describe nanography printing machines features
- 3.2.3 To differentiate nanography and offset printing

## **MODULE IV LENTICULAR PRINTING AND HOLOGRAM**

### **4.1.0 To understand Lenticular printing**

- 4.1.1 To explain Lenticular printing
- 4.1.2 To explain types of Lenticular prints
- 4.1.3 To describe Lenticular lens principle
- 4.1.4 To specify effects of Lenticular print

### **4.2.0 To understand Hologram**

- 4.2.1 To define hologram
- 4.2.2 To explain holography
- 4.2.3 To explain how hologram works
- 4.2.4 To define holography process

### **4.3.0 To know hybrid printing**

- 4.3.1 To define hybrid printing
- 4.3.2 To define hybrid printing machines

## CONTENT DETAILS

### MODULE I

Waterless offset:

Basic principles of waterless offset printing. Plate structure of waterless offset printing. Merits and Demerits of waterless offset printing.

Dry offset:

Principle of dry offset. Dry offset process. Dry offset plates – types, manufacturing methods. Dry offset printing machines. Difference between waterless and dry offset printing.

Print on demand: Print on demand – definition, application, advantages and disadvantages.

### MODULE II

**Textile printing:**

Definition of textile printing. Textile printing steps.

**Methods of printing:** Hand block printing, Perrotine printing, stencil, screen, digital, block printing, burn out printing, blotch , digital, duplex, spray printing , warp printing.

Digital textile printing. Digital printing machines used for textile printing. Screen printing machines used for textile printing.

**Printed Electronics:**

Definition of printed electronics. Advantages of printed electronics. Conventional print processes used in printed electronics. Application of printed electronics.

### MODULE III

**3D printing:**

Introduction about 3D printing. 3D printing process principle. 3D printing technologies - Extrusion deposition, Granular materials binding, Laminated, Photo-polymerization. Steps involved in 3D printing process. Application of 3D printing.

Introduction of (brief) - Bio printing, Nano scale 3D printing.

**Nanography:**

Nanography – definition, inventor. Nanography process. Advantages and disadvantages of nanography. Nano printing machines and its features. Comparison between nanography and offset.

## MODULE IV

### **Lenticular printing :**

Lenticular printing definition. Basic principles of Lenticular printing process. Construction of Lenticular image. Types of Lenticular prints. Principle of Lenticular lens. Lenticular Image and interlacing. Steps involved in Lenticular printing process. Effects of Lenticular print – flip, 3D, animation, zoom and morphing Lenticular

### **Hologram:**

Basic principles of holograms making process. Hologram types. Components of hologram making system - laser, lenses, beam splitter, mirrors, holographic film and Process steps of hologram making system. How hologram works. Requirements for producing hologram – object, laser (object beam, reference beam), recording medium, environment.

**Hybrid printing:** definition of hybrid printing. Introduction to hybrid printing machines. Hybrid printing DI presses. Names of manufacturers and machines for hybrid printing.

| <b>Reference :</b>        |                                     |                                    |
|---------------------------|-------------------------------------|------------------------------------|
| <b>Author</b>             | <b>Title</b>                        | <b>Publishers</b>                  |
| Helmut Kipphan            | Hand book of print media            | Springer Science & Business Media  |
| Bishwanath Chakkaravarthy | Digital colour Printing Technology  |                                    |
| David Bergsland.          | Printing in a digital world         |                                    |
| Christopher Barnatt       | 3D Printing 2 <sup>nd</sup> edition | CreateSpace Independent Publishing |